

Characterization of frictional hysteresis in ball-bearing guideways

F. Al-Bender^{a,*}, W. Symens^b

^a Katholieke Universiteit Leuven, Mechanical Engineering Department, Division PMA, Celestijnenlaan 300B, 3001 Heverlee, Belgium

^b FMTC, Celestijnenlaan 300D, 3001 Heverlee, Belgium

Received 28 April 2004; received in revised form 3 November 2004; accepted 29 November 2004

Available online 29 January 2005

Abstract

Frictional hysteresis is a characteristic that may be found in many machine elements in common engineering use. Plain and rolling element bearings that are widely used in motion guidance of machine tools are typical examples. The study of the non-linear dynamics caused by such elements becomes imperative if we wish to achieve improved design and effective control of such machines.

This paper extends the notion of ‘equivalent’ dynamic quantities, namely stiffness and damping, which describe linear systems to rolling element systems, which exhibit hysteretic friction. The dependency of these quantities on design parameters of the guideway such as preload, ball size and groove angle is furthermore examined experimentally on two set-ups. The first set-up is built to specifically study the hysteretic friction behavior, while the second set-up uses a commercial guideway. This study shows further that a single quantity, the *equivalent damping ratio at resonance*, is able to characterize the dynamics of systems exhibiting hysteretic friction. The greatest part of this paper is therefore dedicated to the experimental correlation of this characteristic damping ratio to the design parameters of rolling element bearings.

© 2004 Published by Elsevier B.V.

Keywords: Rolling elements; Pre-rolling hysteresis; Experimental set-up; Equivalent stiffness; Equivalent damping; Equivalent damping ratio

1. Introduction

Guideways are units that comprise bearings to support the linear motion of machine axes. The most widely used types of bearings for such devices are plane or rolling element bearings. To be able to perform very accurate and small movements with such machine tools, without necessarily lowering the speed, the dynamic characteristics of these guideways have to be known very well. The issue of damping and stiffness of rolling element guideways, especially in the driving direction, has been seldom properly addressed; at least not in proportion with its importance for the positioning and/or machining application. Several researchers analyze the stiffness characteristics of guideways as a function of the load and configuration [1], but do not include damping. Users (machine designers and builders) are not clear about the amount of stiffness and damping that has to be provided

by the guideways (or other parts of the machine for that matter). Such ‘old’ issues are resurfacing nowadays and gaining in importance driven by the increasing demands (of industry) for higher speed and accuracy at a lower cost. One of the possible reasons why these characteristics are not extensively examined in literature is that they originate from highly non-linear contact behavior of which the physics is not clearly understood yet. Complicated models are built to accurately represent the observed phenomena, but physical insight gets lost in such cases. These models are moreover not suitable for motion control design for machine tools with guideways as the parameter sensitivity of these models is quite high and currently used control techniques cannot deal with highly non-linear and sensitive phenomena. It is therefore high time to address these issues more systematically with the aim of understanding better the damping properties of existing guideways, on the one hand, and in order to achieve simple but effective solutions, on the other. This paper aims at identifying *equivalent* quantities to characterize the dynamic behavior of the guideways.

* Corresponding author. Tel.: +32 16 32 25 12; fax: +32 16 32 29 87.
E-mail address: farid.al-bender@mech.kuleuven.ac.be (F. Al-Bender).

One of the non-linear phenomena that hampers a simple characterization of the guideway dynamics is the ‘friction’ behavior between parts that move relative to one another. Much description of friction phenomena can be found in literature, although the nature of the phenomenon is not well understood yet [2,3]. The force resulting from this ‘friction’ can be both velocity and displacement dependent. The friction force dependency on the displacement originates for the phenomenon of microslip [3] and can be described by a rate-independent hysteresis function. Although this displacement dependency is only important for small displacements, up to several tens of microns after a velocity reversal, this motion situation is nevertheless very important in accurate motion systems [4]. It is mainly this effect that determines the stiffness and damping characteristic of the guideway [5]. The present paper therefore considers the pre-rolling hysteresis in isolation from other friction effects which are described in for example [6–8]. The region where the hysteretic friction dominates is called the pre-sliding/pre-rolling region [3,9–12].

Naturally, in order to design controllers for systems exhibiting hysteretic friction, we must be able first to quantify the dynamics of such systems. Second, we must be able to relate the experimentally measured friction hysteresis values to the design parameters of the guideway system (in particular, the ball/roller–groove contact mechanics). This is the topic of this paper. Our basic idea is to extend *equivalent* dynamic quantities, namely, stiffness and damping that describe linear systems to guideways which exhibit the non-linear hysteretic friction phenomenon. The dependency of these properties on design parameters as preload, ball size, surface material and lubrication, and groove angle is then examined experimentally, and correlations are sought.

The greatest part of ‘friction’ losses in rolling element bearings is owing to microslip (except in the rare case of pure rolling when the very small losses will be owing to elastic hysteresis losses in the bulk material.) In commercial recirculating rolling element guideways, there are additional, sliding friction losses at the end seals (the so-called ‘wipers’), which are not considered in this work.

Microslip, in rolling contacts, depends on the ‘creep’ motion, or relative velocity, and on the elasticity, of the points on the opposite surfaces, as they pass through the contact patch. In the case of pure rolling of identical bodies, the creep motion is null, and no microslip takes place. Theoretical quantification of friction losses in rolling contacts is achieved by applying elastic contact theory together with an assumed friction law (usually Amontons’ law, i.e. a constant coefficient of friction) at each contact point [13]. Fig. 1 could give us an idea about the creep motion likely to take place in a rolling ball contact. In Fig. 1a a ball rolls without spin on a plane elastic surface. Because of the different relative surface speeds across the contact patch, the well-known Heathcote slip case occurs. In Fig. 1b, the case of a ball rolling between two V-grooved rails is depicted. The main rolling motion of the ball can be resolved into a rolling and a spinning motion at each contact with the groove surfaces. Consequently, if there

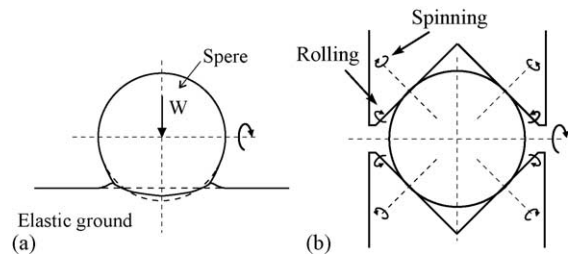


Fig. 1. Elastic sphere rolling over an elastic plane (a) and between two V-grooves (b). In (b) the deformation of the ball and groove at the contact is not shown.

were losses associated with each motion (independently), we would be able to quantify the total friction loss. In particular, we can remark that those losses, and consequently friction, should increase with decreasing groove angle, owing to the increase of spin per unit rolling.

In comparison with the case of steady rolling friction, only very few studies have been carried out on the pre-rolling case. In [14] the relationship between torque and angle of twist for two bodies with an elliptical contact is determined both for the case with no previous stresses and for the case of residual stresses. This relationship is derived using a large number of finite element simulations based on the elastic contact theory of Mindlin. This theory is experimentally verified by Cuttino [15] for a single-nut, preloaded, precision ball screw. The motion of the balls in the set-up consists of a rolling and spinning part, and the spinning part is (simplistically) assumed to be responsible for the hysteretic behavior. The experimental results correspond qualitatively well to the theoretical predictions and quantitative deviations are attributed to a non-uniform loading of the bearing balls and the effects of the wiper seals on the nut, which are not incorporated in the analysis.

Lovell et al. [16–19] performed many experiments with coated ball bearings and different lubricants. In [16] the effect of different lubricants and preloads on the response of balls undergoing an oscillatory motion is analyzed. The transition from dry to lubricant friction and the hysteretic behavior are measured and global tendencies of these phenomena as a function of the preload and viscosity of the lubricant are empirically derived. These experiments are extended to different ball materials or coatings in [17]. In all these experiments the hysteretic behavior is quantified by the steady-state friction torque and rest slope as defined by the Dahl curve [20]. Using elastic Coulomb friction theory these experiments are simulated in a 3-D finite element analysis in [18]. This finite element model is then extended in [19] to predict the influence of preload, coating and lubricant on the frictional behavior of a ball bearing in normal contact between two transversely isotropic coated substrates. From that analysis some empirical laws are derived that describe the dependency of the friction behavior on lubricant, material and preload.

At the Structural Dynamics Lab of the University of Colorado, the friction behavior of precision revolute joints for

space applications is analyzed experimentally. After the analysis of low-speed characteristics they analyzed the pre-sliding behavior using different (dynamical) hysteresis models [21] and tried to combine these with the analytical models of microslip in Hertzian contact that were originally developed by Hinkle et al. [22]. Recently they combined this Mindlin microslip model with the static Maxwell-slip hysteresis model in [23,24] but acknowledged the problem of identifying the parameters of this model. This identification problem stems from the fact that both theories describe different interpretations of the same phenomenon and thus the parameters of both models are interrelated and not independent. They propose to switch between both models as a function of the roughness, that is, the Mindlin model for low roughness and the Maxwell-slip model for high roughness. For this switching a parameter representing the ratio of asperity to bulk deformations as given by Greenwood [25] is used. As outlined by Symens [26] this distinction is however a little artificial as both models can be represented in the same way.

This literature overview shows that a thorough theoretical analysis of the influence of material and geometrical properties on the pre-sliding behavior of sliding and rolling contacts has yet a long way to develop but such theoretical analysis goes beyond the scope of this work. The paper will focus on experimental results that may increase the understanding of the influence of geometrical parameters on the pre-sliding behavior and are based on the initial experiments described in [27,28].

The paper is organized as follows. Section 2 gives a condensed analysis of the dynamic characteristics of a linear mass–spring–damper system. The non-linear analysis of the linear guideways will be based on the concepts defined in this section. Section 3 starts with a comprehensive description of the non-linear hysteretic friction that is present in guideways; thereafter, equivalent damping and stiffness characteristics are derived for this hysteretic friction. Section 4, which forms the bulk of this paper, then validates these theoretical considerations on a test set-up and correlations of the equivalent damping ratio are established with respect to bearing design parameters. Finally some conclusions and directions for further research are highlighted in Section 5.

2. Equivalent damping ratio of a linear mass–spring–damper system

A linear guideway system allows one degree of freedom (DOF), the feed direction (X), and ‘blocks’ the remaining five, the bearing DOFs (normal (X), lateral (Y), yaw (M_C), pitch (M_B), and roll (M_A)) (see Fig. 2). Such a linear guideway can be modeled as a mass moving in one direction with friction forces between the mass and the fixed ground.

If the system is linear, the problem reduces to considering a single DOF (SDOF) mass–damper–spring (m – c – k) system in

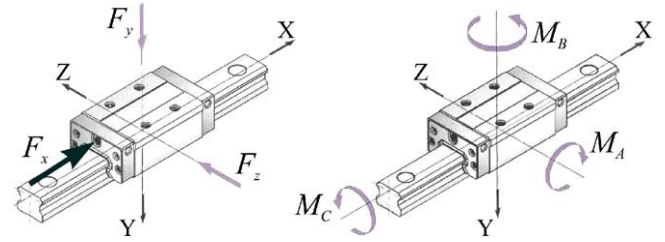


Fig. 2. Configuration scheme of a linear guideway.

the drive direction. This section briefly summarizes the properties of a linear mass–spring–damper system as described in [29]. The concepts derived here are extended to systems exhibiting hysteretic friction in the next section.

The equation of motion of an SDOF m – c – k system is given by:

$$m\ddot{x} + c\dot{x} + kx = F(t) \quad (1)$$

with F the external (dynamic) force, x the displacement in the X -direction and dots representing derivatives with respect to time. Defining $\omega_n = \sqrt{k/m}$, the *natural frequency*, and $\zeta = c/2\sqrt{km}$ the *damping ratio*, Eq. (1) can be written as:

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{F}{m} \quad (2)$$

Further, normalizing the time and the natural frequency by $\tau = \omega_n t$ and $\bar{\omega} = \omega/\omega_n$, Eq. (2) becomes:

$$x'' + 2\zeta x' + x = \frac{F}{m\omega_n^2} = \bar{F} \quad (3)$$

with accents denoting derivatives with respect to τ . This normalized form of the equation of motion of a linear second order system is very significant in that it shows that there is only one single parameter that determines the dynamic behavior of the system; namely, the equivalent damping ratio ζ . In a linear system, ζ is constant. In a ‘linearized’ equivalent of a non-linear system, ζ will generally depend on the oscillation amplitude and/or frequency. This ‘equivalent’ ζ will be analyzed for systems with hysteretic friction in the next section.

3. Equivalent damping ratio of a mass on a hysteretic spring

This section first briefly characterizes the hysteretic friction that is dealt with in this paper. Hereafter the concepts introduced in the previous section are extended to systems exhibiting hysteretic friction.

3.1. Analysis of hysteretic friction

With hysteretic friction is meant a friction force that is described by a displacement dependent hysteresis function.

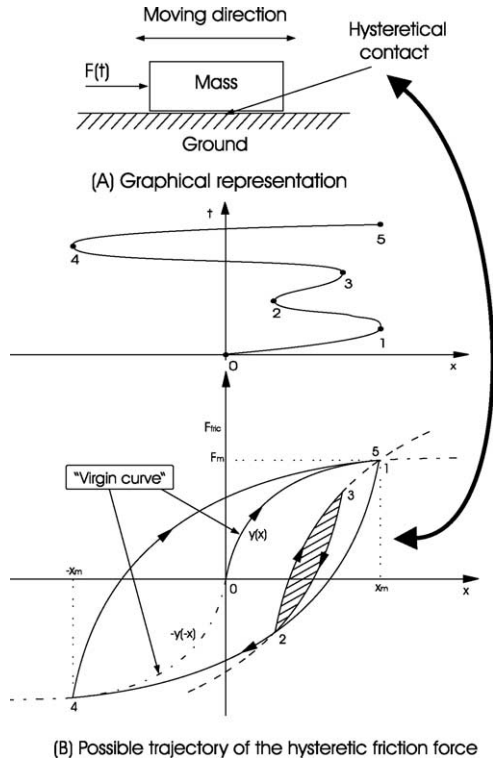


Fig. 3. Mass subject to 'hysteretic friction'. A graphical interpretation of the set-up under consideration is given in (A). An example of a hysteresis trajectory is given in (B). Based on the predefined position trajectory the friction force is obtained as follows: the motion starts at point (0,0) along $y(x)$, reverses at point 1 and follows (1–2–4). After reversal at point 2, curve (2–3–5) is followed; after reversing at point 3, the force curve follows (3–2) and again the (1–2–4) line to arrive at 4, points 2 and 3 can now be removed from the memory.

Fig. 3B shows the time trajectory of the friction force resulting from an imposed displacement trajectory.

The main characteristics of this friction force are:

- The force is displacement dependent and not velocity dependent.
- There is no unique instantaneous relation between the friction force and the displacement: the force also depends on the traversed trajectory history.
- The states of the system (position and velocity) are not sufficient to describe the friction force. The velocity reversal points have also to be remembered.
- If certain memory points (velocity reversals) are revisited they can be wiped out of the memory. This is called the closing of an internal loop [30].
- The first derivative of the friction force is not continuous at velocity reversals.

The mathematical description of the hysteretic friction is based on two rules (Eqs. (4) and (5)). The second rule is applied most of the time since the first rule (Eq. (4)) only applies if the friction object enters a region where it has never entered before. The parameters F_m and x_m in Eq. (5) change value at each velocity reversal point, or when an internal loop

is closed [31,32].

$$\text{Rule I : } \begin{cases} F_{fric} = f(x) \\ \text{with } f(x) = \begin{cases} y(x) & x \geq 0 \\ -y(-x) & x \leq 0 \end{cases} \end{cases} \quad (4)$$

$$\text{Rule II : } \begin{cases} F_{fric} = F_m + 2f\left(\frac{x - x_m}{2}\right) \\ F_m = F_{fric}(x_m) \text{ calculated with the} \\ \text{formula for } F_{fric} \\ \text{before the reversal} \end{cases} \quad (5)$$

Both rules are completely characterized by the virgin curve $y(x)$ of the hysteretic friction. This curve starts in general at the origin, has a positive first and a negative second derivative for all x , and saturates for large x . A more detailed description of the behavior of hysteretic friction can be found in [31,5].

This type of hysteresis can be simulated by the so-called Maxwell-slip model that consists of a set of Coulomb friction blocks connected, in parallel, via linear springs. It is characterized by its 'memory' effect and by its rate (i.e. frequency) independence [33]. The dynamic equation of the system shown in Fig. 3A can then be written as:

$$m\ddot{x} + \phi(x, \text{his}(x_m^i)) = F(t) \quad (6)$$

with $\phi(x, \text{his}(x_m^i))$ representing the hysteretic friction force. In this formula $\text{his}(x_m^i)$ represents all the history of the traversed movement that is relevant for the future movement and thus has to be kept in the memory.

3.2. Equivalent stiffness, damping and damping ratio

The highly non-linear dynamic equation describing the motion of a mass subject to hysteretic friction (Eq. (6)) can be solved analytically only for the free response case, where $F(t) = 0$ [5]. In the general case, where $F(t) \neq 0$, it is not possible to derive an analytical solution. To investigate this equation some approximating methods can be used, which are relevant to periodic motion. One of these is the describing function method where the non-linear element $\phi(x, \text{his}(x_m^i))$ is replaced by an element that gives as output the fundamental component, in Fourier terms, of the output of the non-linear element for a sinusoidal input. Doing this, the system is 'linearized' in a harmonic way. This method is used in [5,31,32] to analyze the dynamics of a mass subject to hysteretic friction and will be used in this section to obtain equivalent damping and stiffness characteristics of guideways. Fig. 4 shows the steady-state hysteresis loops that are considered in the analysis.

The non-linear element in Eq. (6) is represented, for periodic motion, by $\phi(x, A_m)$, where A_m is the amplitude of the motion. The general formulation of the Fourier expansion is

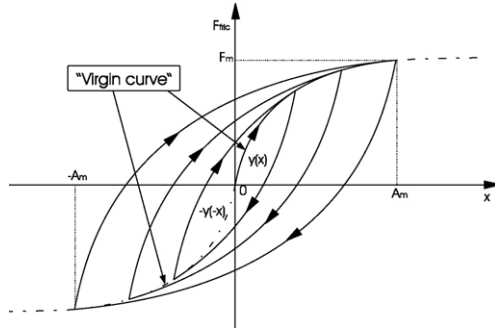


Fig. 4. Hysteresis loops under sinusoidal excitation.

given by:

$$\phi(x, A_m) = \sum_{n=0}^{\infty} a_n \cos(\omega_n x) + \sum_{n=1}^{\infty} b_n \sin(\omega_n x) \quad (7)$$

$$\text{with } \begin{cases} \omega_n = n\omega_0, \omega_0 = \frac{\pi}{A_m} \\ a_0 = \frac{1}{2A_m} \int_{-A_m}^{A_m} \phi(x) dx \\ a_n = \frac{1}{A_m} \int_{-A_m}^{A_m} \phi(x) \cos(\omega_n x) dx \\ b_n = \frac{1}{A_m} \int_{-A_m}^{A_m} \phi(x) \sin(\omega_n x) dx \end{cases} \quad (8)$$

In order to calculate the describing function, the fundamental Fourier components (a_0 , a_1 and b_1 from Eq. (7)) of $\phi(x)$ need to be evaluated. Assume therefore that $x = A_m \cos \theta$ with $0 \leq \theta < 2\pi$, that is $-A_m \leq x \leq A_m$. Then:

$$\phi(x) = \begin{cases} -y(A_m) + 2y\left(\frac{A_m + x}{2}\right) = \phi^+ & -\pi \leq \theta \leq 0 \\ -y(A_m) - 2y\left(\frac{A_m + x}{2}\right) = \phi^- & 0 \leq \theta \leq \pi \end{cases} \quad (8)$$

yielding for the fundamental components of the friction force for a cosinusoidal displacement input:

$$\begin{aligned} a_0 &= 0 \\ a_1 &= -\frac{4}{\pi} \int_0^{\pi} y\left(\frac{A_m}{2}(1 - \cos \theta)\right) d\theta \\ b_1 &= -\frac{1}{\pi A_m} \left\{ 8 \int_0^{A_m} \left(y(x) - \frac{y(A_m)}{2} \right) dx \right\} \end{aligned} \quad (9)$$

The fundamental Fourier terms can be physically interpreted as follows:

- The fact that $a_0 = 0$ indicates that there is no DC shift.
- The term between $\{\dots\}$ in the expression of b_1 in Eq. (9) is equal to the area enclosed by the hysteresis loop of amplitude A_m .
- Since the describing function linearizes the system in the way described above, it is possible to extract an equivalent stiffness and damping from the fundamental Fourier terms. The equivalent stiffness of the hysteresis, which is related

to the displacement, is given by: $k_e(A_m) = a_1/A_m$, and can be approximated by $k_e \approx F_m/A_m$, i.e. the slope of the 'major axis' of the hysteresis loop. It achieves its maximum value as the displacement approaches zero, and its minimum value ≈ 0 , as the displacement approaches infinity. Although this value can be used for a qualitative analysis care has to be taken when using this approximation for a quantitative analysis (in a certain region of the frequency response function) as is shown in [5,31,32].

- The equivalent damping, which is related to the velocity, is given by: $c_e(A_m)\omega = -b_1/A_m = \text{loop area}/\pi A_m^2$ i.e. the damping force is equal to $c_e(A_m)\omega A_m = \text{loop area}/\pi A_m$. Thus this amount of energy dissipation per cycle does not depend on the velocity, as for viscous damping, but depends on the amplitude of the motion.

Using $k_e(A_m)$ and $c_e(A_m)$, Eq. (6) can be approximated by:

$$m\ddot{x} + c_e(A_m)\dot{x} + k_e(A_m)x = F_0 \sin(\omega t + \beta) \quad (10)$$

In this formulation the phase difference between the force and displacement is represented by β in the force term. The resulting effect of the interplay between the two varying quantities $c_e(A_m)$ and $k_e(A_m)$ on the dynamic behavior as a function of the excitation amplitude is hard to interpret, as shown by several experimental measurements [34,35]. The parameters are moreover very sensitive with respect to system properties such as preload, materials used and the geometry of the contact surface. The reason for the difficulty in interpretation is that the dynamics of a single-degree-of-freedom mass-spring-damper system is determined by one (dimensionless) parameter only, the equivalent damping ratio, as shown in the previous section.

The results of the previous section for a linear mass-spring-damper system can thus readily be extended to the hysteretic spring case giving rise to an excitation amplitude dependent equivalent damping ratio:

$$\begin{aligned} \zeta(\omega, A_m) &= \frac{c_e(A_m)}{\sqrt{k_e(A_m)m}} = \frac{c_e(A_m)\omega_n(A_m)}{2k_e(A_m)} \\ &= \frac{c_e(A_m)\omega(A_m)}{2k_e(A_m)} \left(\frac{\omega_n(A_m)}{\omega(A_m)} \right) \end{aligned} \quad (11)$$

In order to eliminate the frequency dependency, only the damping ratio at resonance ($\omega = \omega_n$) is considered as a general measure of the dynamics [27]:

$$\zeta_r(A_m) = \frac{c_e(A_m)\omega_n(A_m)}{2k_e(A_m)} \quad (12)$$

Substituting $c_e = \text{loop area}/\pi A_m$ and approximating k_e by $k_e \approx F_m/A_m$, i.e. the slope of the 'major axis' of the hysteresis loop, we obtain:

$$\zeta_r(A_m) = \frac{\text{loop area}}{2\pi F_m A_m} \quad (13)$$

which is independent of the resonance frequency but depends on the vibration amplitude A_m . This parameter can give us

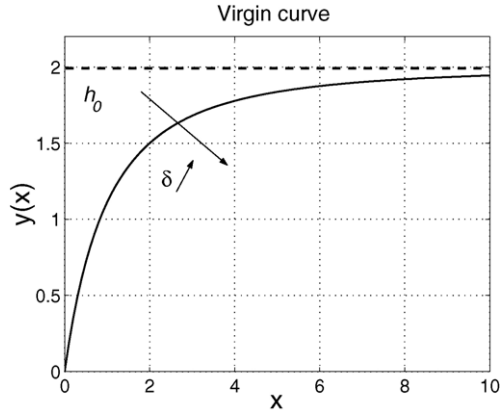


Fig. 5. Virgin curve for Eq. (14).

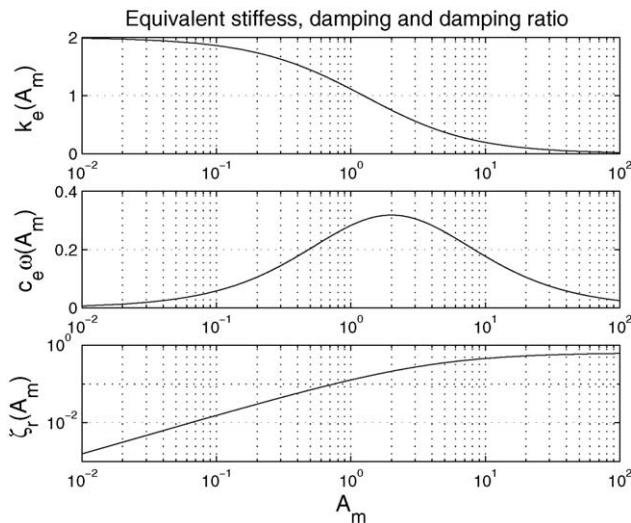
a good idea about the damping behavior in general. Beside knowing the value at resonance, we also know that it will be higher, by the ratio ω_n/ω at lower frequencies (and vice versa at higher frequencies). We see thus that the equivalent damping ratio depends only on the shape of the hysteresis curve. It is independent of the mass and the magnitude of the stiffness.

The hysteretic behavior can be calculated theoretically once the hysteresis virgin curve is known. As a ‘fictitious’ (theoretical) example of a simple virgin curve, consider the function:

$$f(x) = h_0 \left[1 - \left(\frac{\delta}{x + \delta} \right)^2 \right] \quad (14)$$

where the parameters h_0 and δ are scaling factors for the force and displacement respectively (see Fig. 5).

Using this representation of the virgin curve, the equivalent stiffness, damping and damping ratio become

Fig. 6. Equivalent stiffness, damping and damping ratio for the virgin curve of Eq. (14) with $\delta = 2$ and $h_0 = 2$. All variables have arbitrary units.

(see Fig. 6):

$$k_e = \frac{h_0}{A_m} \left[1 - \left(\frac{\delta}{A_m + \delta} \right)^2 \right] \quad (15)$$

$$c_e \omega = \frac{4h_0}{\pi} \frac{A_m}{(A_m + \delta)^2} \quad (16)$$

$$\zeta_r = \frac{2}{\pi} \frac{A_m^2}{A_m^2 + 2A_m\delta} \quad (17)$$

The stiffness approaches a constant value, h_0 , for small displacement and $1/A_m$ for large displacements. In between these two extremes, the form depends on the hysteresis shape. ζ_r is independent of F_m , goes to zero for A_m approaching zero and approaches a constant value for A_m approaching infinity. These observations however do not hold universally. If the virgin curve for example can be written as a power n of x , with $0 < n < 1$, ζ_r will be constant and equal to $(1 - n)/(1 + n)$. The question now arises as to whether:

- it is possible to identify the contact parameters influencing the shape of ζ_r ; e.g. the roughness form, the osculation form, surface materials, lubricant, etc.,
- one is able to ‘engineer’ the shape of the curve, via playing on one or a combination of such parameters, to the end of achieving a certain desired damping behavior.

A theoretical derivation of this parameter dependency is out of the scope of this work, but several experiments were performed to investigate this dependency. The results of these experiments are presented next.

All quantities defined above depend on the specific form of the virgin curve of the hysteresis. This form, and thus also the equivalent stiffness and damping (ratio), depends on material, preload, geometric properties, etc. of the contact. To the knowledge of the authors only few and very specific results can be found in literature that analyze these correlations (see Section 1).

4. Experimental measurement of the equivalent damping ratio at resonance

Two set-ups are used to analyze the hysteretic friction in linear guideways experimentally. The first set-up, which is specially designed to analyze the influence of hysteresis on the dynamical behavior of machine tool axes, is used to investigate the influence of different system properties on the equivalent damping ratio at resonance ζ_r . A second set-up, which consists of a commercial guideway, is used to analyze the variation of the characteristics over a large working range. The properties that are analyzed are the preload (W), the materials used (mat), the groove angle of the guideway (α), the ball diameter (R) and the linear rolling displacement of the guideway (X). The aim is to experimentally reveal the

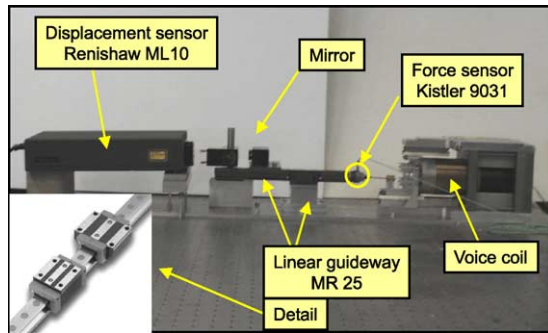


Fig. 7. Set-up with a commercial linear roller guideway (Schneeberger MR25 G2-A-02-09). The carriage is fixed, while the rail is driven by a linear voice-coil. The position of this rail is measured with an optical laser encoder.

dependency of the virgin curve on these parameters, that is, to find the function:

$$F_{\text{fric}} = \psi(x, W, \text{mat}, \alpha, R, X). \quad (18)$$

The set-ups are described first, after which the experimental results are presented and discussed.

4.1. Description of the test set-ups

4.1.1. Set-up with commercial guideway

Fig. 7 shows the set-up consisting of a commercial linear roller guideway (Schneeberger MR25 G2-A-02-09). In this set-up the carriage is fixed, while the rail is driven by a linear coil motor. The guideway does not consist of balls, but of rollers. The position of this rail is measured with an optical laser encoder with a resolution of 10 nm (Renishaw ML10).

4.1.2. Set-up with roller guideway

Figs. 8 and 9 show the set-up consisting of two V-grooved rails between which two balls roll. One rail is fixed, while the other is actuated in the sliding direction by a linear exciter (Brüel & Kjær 4810). The force on the guideway is measured by a force sensor (Kistler-GEPA 9031). The relative displacement between the two guideways is measured by an eddy-current displacement sensor with a resolution better than 0.1 μm (Bently type 3300). Note that both force is applied and displacement is measured at the centerline of the rolling balls. As Fig. 8 shows, there is also a possibility to vary the preload on the guideway. Rails of different geometry (V-angle) can be mounted and different balls (size and material) can be inserted in the set-up. In this way, the influence of these variables on the hysteretic friction can be measured and classified.

4.2. Experimental results

In all experiments, the steady-state response of the system to very low frequency sinusoidal input forces with a certain amplitude is measured. These steady-state responses are periodic and are called ‘hysteresis loops’. To reduce the influence

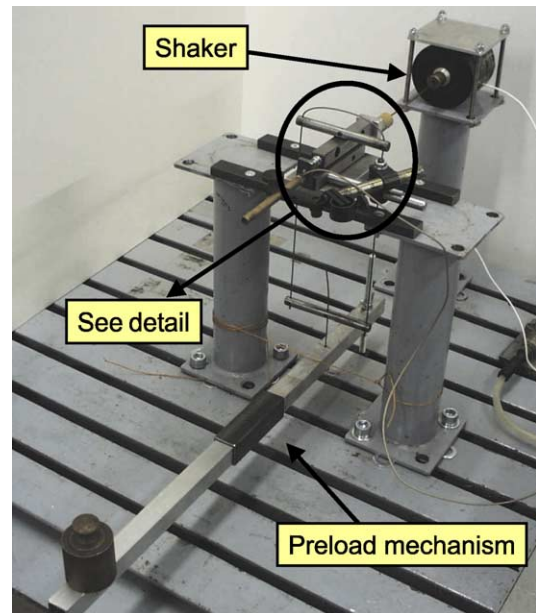


Fig. 8. Set-up consisting of two V-grooved guideways between which two balls roll. One guideway is fixed, while the other is actuated with a linear actuator.

of noise, the average of 10 measured loops, for each run, is taken for the analysis. From the measured hysteresis loops the quantities that are defined above can be derived. These loops depend on the amplitude of the applied force, which determines the displacement amplitude. For each configuration the applied force was increased until rolling commenced. The values of the characteristic parameters were calculated for each case.

The equivalent stiffness and damping are not displayed separately as they qualitatively agree with the theoretical prediction of Fig. 6 and their quantitative values are of minor importance for the presented analysis.

4.3. Set-up with commercial guideway

First, the results from the commercial guideway are detailed. Fig. 10 shows several hysteresis loops with different amplitudes for the commercial set-up at one position X . The figure also shows the identified virgin curve of the form represented by Eq. (14) for these loops. This virgin curve fits the

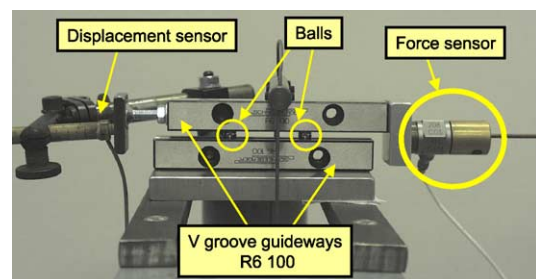


Fig. 9. Detail of the set-up consisting of two V-grooved guideways between which two balls roll.

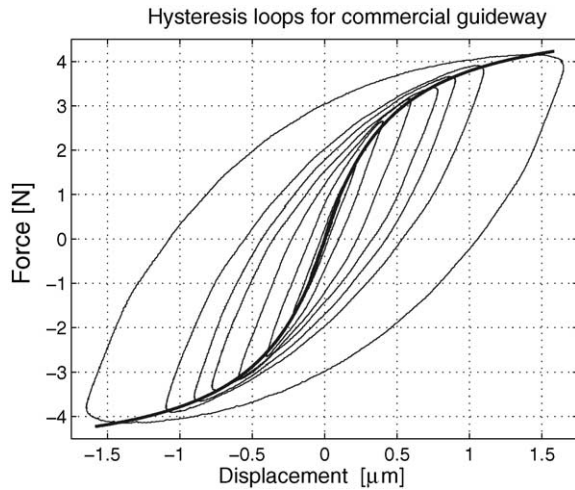


Fig. 10. Measured hysteresis loops with identified virgin curve for the commercial guideway.

loops quite well and therefore the form is suitable to describe the hysteresis in these guideways.

In Fig. 11a the same kind of hysteresis loops are shown for different positions of the guideway on the rail (the guideway is moved manually between the different experiments). From

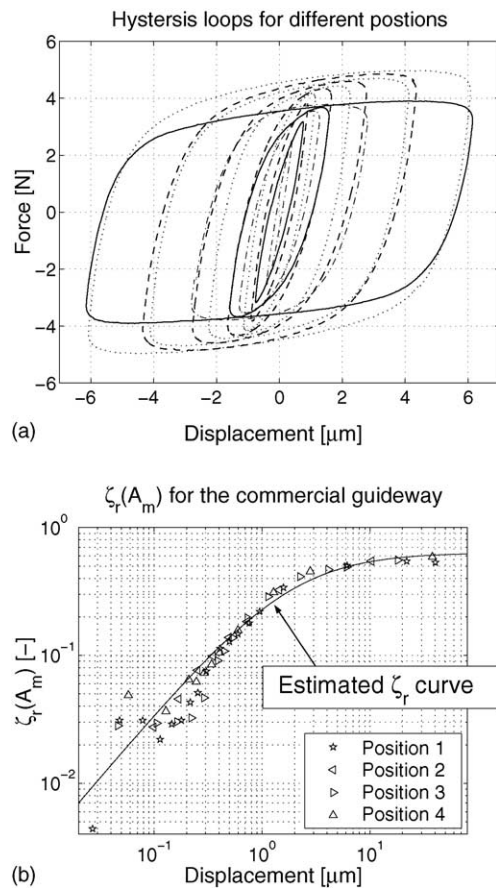


Fig. 11. Experimental results for four different positions of the commercial guideway. (a) Hysteresis loops for different positions (solid, dashed, dotted and dashed-dotted line). (b) ζ_r -curves for different positions.

this figure it is clear that the form of the loops is similar for all the positions, but the exact values of the parameters δ and h_0 , for a virgin curve modeled according to Eq. (14), differ significantly. This position dependency, which is primarily caused by the change of preload owing to the guideway carriages being form-closed, poses a problem if the guideway is to be modeled or characterized accurately based on the parameters δ and h_0 .

The equivalent damping ratio at resonance ζ_r , calculated for the different positions of the guideway following Eq. (17), is shown in Fig. 11b. In contrast to the hysteresis loops, the equivalent damping ratio at resonance shows little sensitivity to the position of the guideway. This can be explained by (i) the fact that ζ_r is independent of the preload (see Section 4.4.2) and (ii) the averaging effect of the calculation of the area of the hysteresis loop in the formula of ζ_r (Eq. (13)). The small spread of the data for small displacements can furthermore be attributed mainly to the low resolution of the force measurements. The figure also shows the theoretical ζ_r -curve of Eq. (17) based on the identified virgin curve in Fig. 10. This curve corresponds very well to the ζ_r values calculated from the measured hysteresis curves.

Based on these initial experiments, the equivalent damping ratio at resonance ζ_r seems to be a good global characteristic parameter to describe the guideway as it is position-independent, in contrast to a detailed identification of the virgin curve, and describes the dynamics in one lumped quantity.

4.4. Set-up with roller guideway

The specially designed set-up of Fig. 8 is used to analyze the dependency of the hysteresis characteristics and the equivalent damping ratio at resonance on material, load and geometric parameters of the guideway (see Table 1).

The results on this set-up are expressed in terms of a single ball, so that global results of a guideway system can be deduced in function of the number, size, orientation, etc. of the balls. Multiple balls can be handled as a parallel connection of single balls. (Note that ζ_r is independent of the number of balls).

4.4.1. Different materials

Fig. 12a shows hysteresis loops for a set-up with a V-groove angle of 90° , a preload of 54 N per ball, a ball diameter of 6 mm and three different types of ball materials: steel, Si_3N_4 and steel with a diamond coating. Only one loop is shown for each of the materials for clarity, although several have been measured [28]. The presented loops are the ones

Table 1

Different test parameters for the hysteresis characteristics and the equivalent damping ratio at resonance

Parameter	Values
Material (mat)	Steel, diamond-coated steel, Si_3N_4
Load (W)	10–55 N
Groove half angle (α)	30° , 60° , 90°
Ball diameter (R)	1.5 and 6 mm

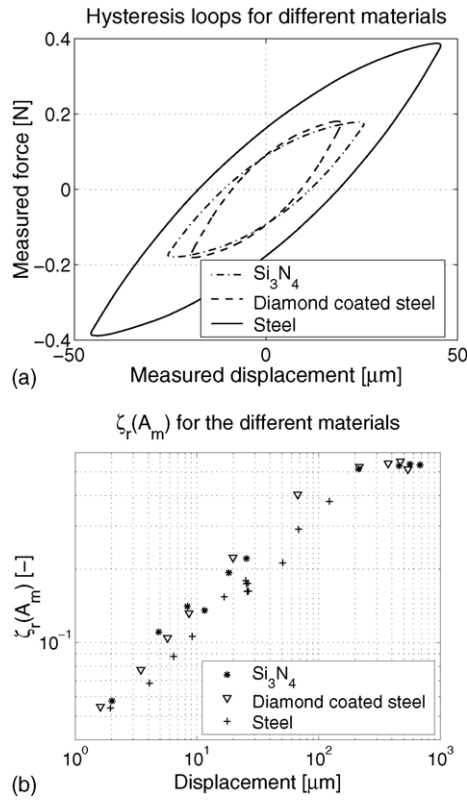


Fig. 12. Experimental results for the roller guideway set-up with steel, Si_3N_4 and with diamond coated steel balls. (a) Hysteresis loops. (b) ζ_r -curves.

obtained just before rolling occurs. The figure clearly shows that the pre-rolling-distance is the smallest for the diamond coated steel balls, followed by the Si_3N_4 balls. The biggest distance is obtained for the steel balls. The rolling force is the largest for the steel balls and is approximately the same for the diamond coated balls and the Si_3N_4 balls. It is not the aim of this work to explain this difference, but to give a methodology to measure the different behaviors. Nevertheless, we can offer the following explanation for this phenomenon. The pre-rolling distance (like the pre-sliding distance) will decrease with increasing tangential stiffness (per unit contact area) and decreasing adhesion of the contacting surfaces (and near surface material). An examination of the Maxwell-slip model will reveal this fact, which is consistent with the experimental results obtained, namely that the harder (therefore stiffer) materials show smaller pre-rolling distance [9].

The corresponding ζ_r -curves are shown in Fig. 12b. From this figure, it can be concluded that the (small) difference in the hysteresis loops for the diamond coated steel balls and the Si_3N_4 balls are hardly detectable in the ζ_r -curves for these two ball materials. The ζ_r values of the guideway with the steel balls are approximately 22% lower than those of the two other ball materials.

4.4.2. Various loads

To analyze the influence of the preload of the guideway on the hysteresis characteristics experiments are conducted

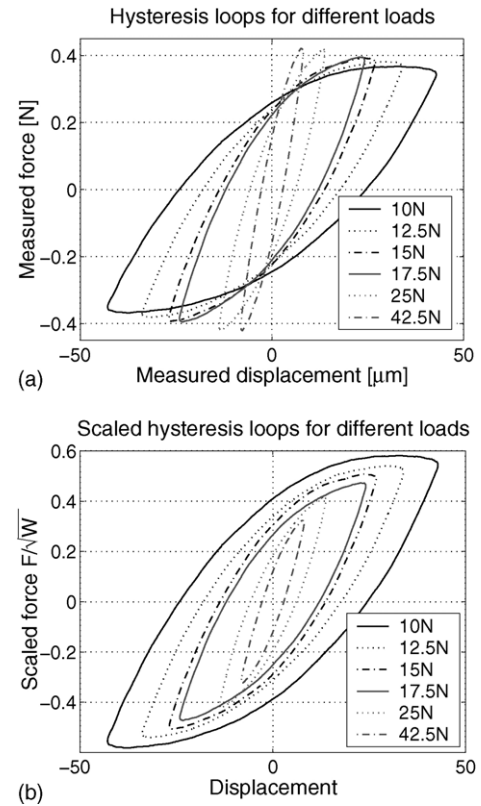


Fig. 13. Hysteresis loops for the roller guideway set-up with steel balls of 6 mm diameter, a groove angle of 90° and various loads. (a) Original loops. (b) Scaled loops.

with different weights on the cantilever beam of Fig. 8. The load for one ball is derived as: (equivalent mass of the cantilever + added mass) $\times 5 \times 9.81/2$, with 2 being the number of balls. Fig. 13a shows hysteresis loops for several loads of the configuration with steel balls with a diameter of 6 mm and a groove angle of 90° . Although a large number of load increments has been used only a set is displayed to improve the clarity of the figure. Increasing the normal load leads to increasing the contact area (real and apparent) and therefore the tangential stiffness of the contact. Hence, the initial slope of the hysteresis curve, which is proportional to the tangential stiffness, increases with load. If the friction force F is now scaled as $F/\sqrt{W}$ the hysteresis loops will approximately coincide with those corresponding to one and the same virgin curve (cf. Fig. 10), as shown on one macroscopic position of the guideway as shown in Fig. 13b. The choice of this type of scaling is motivated by elastic contact theory considerations. The transformation is not exact and a better scaling in the form of F/W^n , with n optimized, can most probably be obtained. The number of conducted experiments is however not sufficiently large to obtain a better value for n with a high significance.

The ζ_r -curves for the different load values approximately coincide, as would be expected (see Fig. 14). A possible explanation for the small differences is that there is a slight influence of the position dependency of the friction behavior and, again, the insufficient force measurement resolution

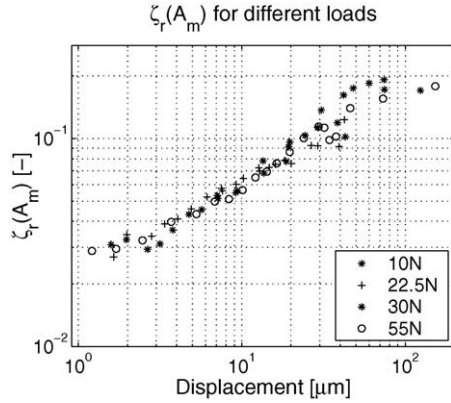


Fig. 14. ζ_r -curves for the roller guideway set-up with steel balls of 6 mm diameter, a groove angle of 90° and various loads. Only the results for a number of loads is shown to increase the clarity of the figure.

for small displacements. The different positions result from the nature of the experiments: for every load, the excitation amplitude is increased until rolling occurs; after rolling, the rest position of the balls always differs from the initial rest position.

Furthermore, we can read, as typical values for design purposes, that the damping ratio will have a value of about 6% at a displacement amplitude of $10\ \mu\text{m}$ and about 3% at $2\ \mu\text{m}$. By extrapolation, the value at $1\ \mu\text{m}$ is expected to be approximately 2.1%; this has to be verified by dedicated tests, however, since it is dangerous to deduce such values by extrapolation.

Similar observations were made for the steel balls with a diameter of 1.5 mm and a groove angle of 90° and for the diamond coated and Si_3N_4 balls with a diameter of 6 mm and the three different groove angles.

These measurements are not shown as they do not reveal any new information; they can be found in [28].

4.4.3. Geometric parameters: groove angle and ball diameter

To investigate the influence of the groove angle the experiments are repeated for three groove angles ($\alpha = 60^\circ$, 90° and 120°) using steel balls with a diameter of 6 mm. The groove angle governs the ratio of rolling to spinning, so that the smaller the groove angle, the higher the resulting damping will be. Fig. 15a shows several hysteresis loops for each angle so that the influence of the groove angle is clearly visible. Again a scaling factor is looked for that represents the dependency of the virgin curve on α . Fig. 15b reveals that when the force is scaled as $F \sin^2(\alpha/2)$ and the displacement as $x/\sin^2(\alpha/2)$ the hysteresis loops all coincide approximately. The motivation for this choice of scaling is the $\sin^2$ rule for stiffness. As in the case of the load dependency, this scaling is not necessarily optimal, and more experiments have to be conducted to obtain an empirical law with high significance.

The ζ_r -curves for different groove angles are shown in Fig. 16a. These curves clearly do not coincide, but they do after scaling, as shown in Fig. 16b.

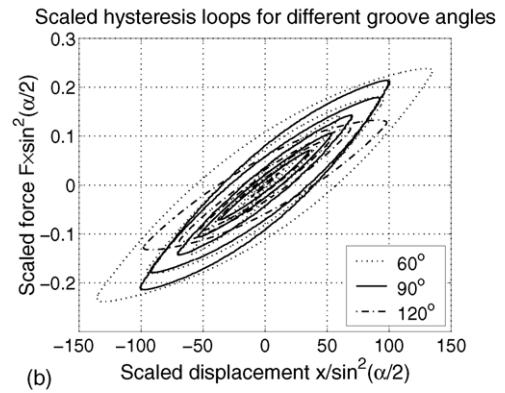
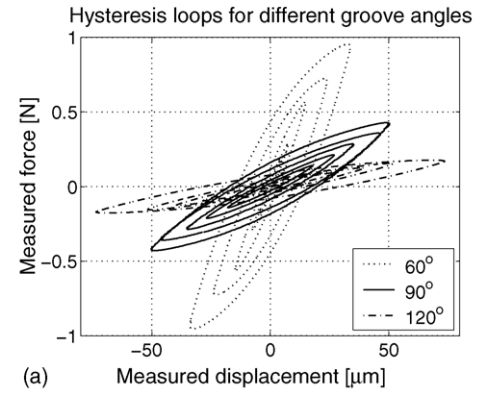


Fig. 15. Hysteresis loops for the roller guideway set-up with steel balls of 6 mm diameter and three groove angles: 60° , 90° and 120° . (a) Original loops. (b) Scaled loops.

For the set-up with the roller guideway, steel balls with a diameter of 6 mm and with a diameter of 1.5 mm could be used. Fig. 17a shows the hysteresis loops for a normalized load for the two diameters (and $\alpha = 90^\circ$). If the displacement x is scaled as $x/\sqrt{R}$ (motivated by elastic contact theory) the hysteresis loops are transformed to those shown in Fig. 17b. The figure shows that the transformation captures the dependency of the virgin curve on R approximately. In Fig. 18a the (non-coinciding) ζ_r -curves for the different ball diameters are presented while Fig. 18b shows the (better coinciding) scaled ζ_r -curves.

For both experiments, the remaining differences between the loops can again be explained by the position dependency of the hysteresis loops. The different positions result from the mounting and demounting that was necessary to exchange the guideways with different groove angles and the balls with different diameters.

4.5. Discussion

Different experiments have been performed to characterize hysteresis damping in ball–groove guideways and to obtain empirical laws describing the influence of various system properties on the parameters of the virgin curve and the curves that quantify the equivalent damping ratio at resonance as a function of the displacement amplitude.

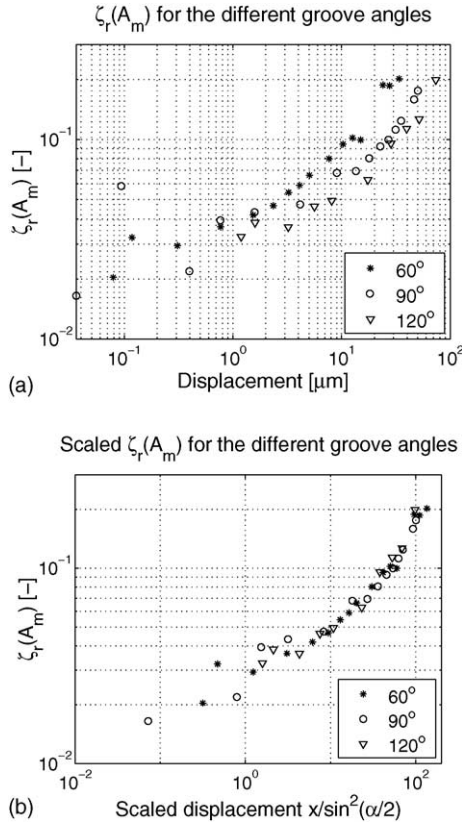


Fig. 16. ζ_r -curves for the roller guideway set-up with steel balls of 6 mm diameter and three groove angles: 60°, 90° and 120°. (a) Original curves. (b) Scaled curves.

The experiments on the set-up with the commercial guideway clearly showed that the parameters describing the virgin curve of the hysteretic friction are very sensitive to the (macroscopic) position of the guideway, although the (qualitative) form itself remains the same. The equivalent damping ratio at resonance on the other hand does not suffer from this position dependency since we have seen that the ζ_r -curves are almost independent from preload.

The measurements on the roller guideway set-up revealed that the hysteresis characteristic significantly depends on the materials used for the balls in the guideways. In particular the damping ratio for the Si₃N₄ and diamond coated steel balls is higher than that for the steel balls. Further experiments also showed that the normal force (or preload) scales the force axis of the hysteresis curve approximately as $F/\sqrt{W}$. Increasing the normal force thus leads to an increase in the stiffness and damping simultaneously (see Section 4.4.2), for a given displacement. Since the equivalent stiffness and damping will increase by the same ratio, the equivalent damping ratio at resonance will not be effected by the normal force. Experiments with different groove angles showed that this parameter scales the axis of the virgin curve as $F \sin^2(\alpha/2)$ and $x/\sin^2(\alpha/2)$ respectively. Experiments with different ball diameters finally showed that the ball diameter approximately scales the displacement axis of the hysteretic friction as $x/\sqrt{R}$.

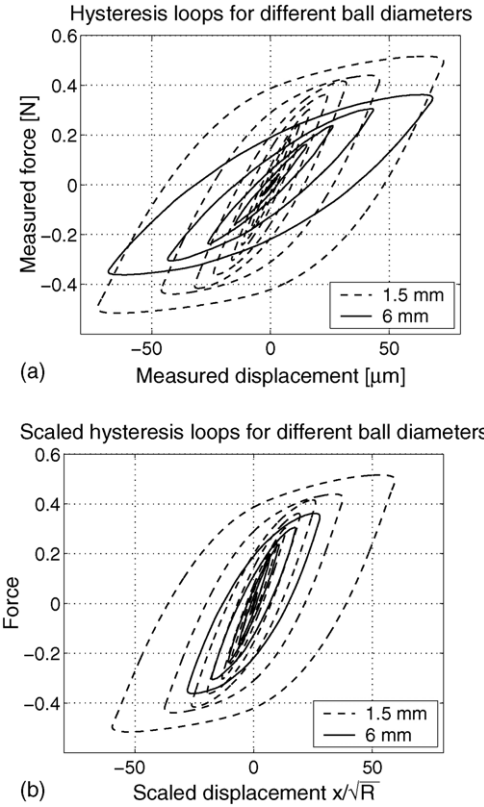


Fig. 17. Hysteresis loops for the roller guideway set-up with a groove angle of 90° and steel balls with a diameter of 1.5 and 6 mm for different loads. (a) Original loops. (b) Scaled loops.

Based on the performed experiments the relation between the hysteretic friction force and the different system parameters can be written as:

$$\frac{F_{\text{fric}} \sin^2(\alpha/2)}{\sqrt{W}} = \tilde{\psi} \left(\frac{x}{\sqrt{R} \sin^2(\alpha/2)}, \text{mat}, X \right) \quad (19)$$

Although these scale laws may be (rough) approximations, they could provide useful guidelines for the design and selection of rolling element guideways when damping (in the rolling direction) is of importance.

From the performed experiments it can be concluded that the parameters of the virgin curve are highly sensitive to small variations of the system properties. The equivalent damping ratio on the other hand is less sensitive to the value of these properties. As highly sensitive parameters are practically hard to identify and of little use in system analysis and control synthesis, the equivalent damping ratio proves to be a good measure to characterize and classify the dynamics of guideways exhibiting hysteretic friction. It moreover captures the system dynamics in one (dimensionless) quantity. Therefore this parameter is ideal for use in designing motion controllers for systems with hysteretic friction. For more details we refer the interested reader to [26].

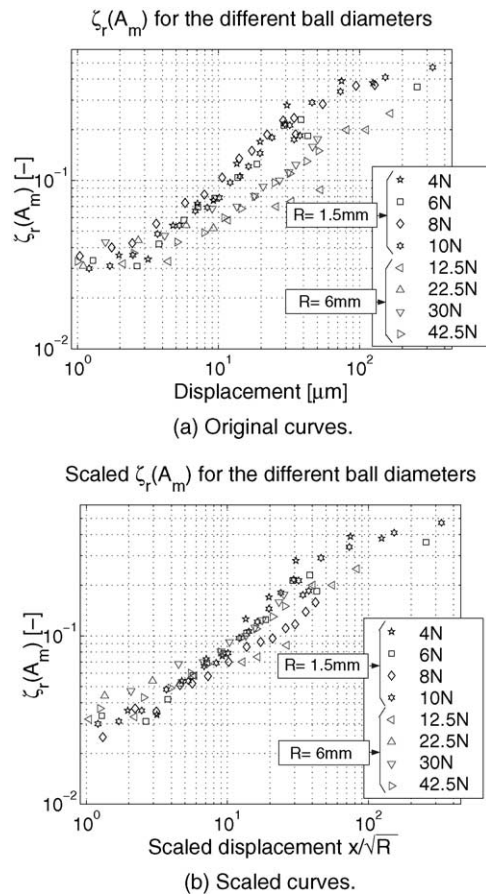


Fig. 18. ζ_r -curves for the roller guideway set-up with a groove angle of 90° and steel balls with a diameter of 1.5 mm (black line) and 6 mm (gray line) for different loads. (a) Original curves. (b) Scaled curves.

5. Conclusions

This paper has considered the dynamic behavior of single-degree-of-freedom systems consisting of a mass on a non-linear ‘hysteretic spring’. To describe the dynamics of such systems adequately ‘equivalent’ dynamic quantities have been employed analogous to those used for describing linear dynamical systems, except for their amplitude dependence. The dependency of these parameters on several guideway properties such as preload, ball size, groove angle and used materials has furthermore been investigated experimentally. This analysis shows that one parameter, the equivalent damping ratio at resonance, is able to robustly describe the dynamics of guideways exhibiting hysteretic friction. This parameter therefore proves to be a good measure to characterize and classify the dynamics of such guideways.

Finally, because of the importance of hysteretic friction in the dynamics of mechanical positioning systems, this non-linear phenomenon should not be overlooked, as is hitherto the case, in the control design for such systems. In future research, therefore, ways to use the equivalent damping ratio at resonance to design (non-linear) controllers which outperform the currently used linear controllers will be considered.

Acknowledgments

This research is sponsored by the Belgian program of Interuniversity Poles of Attraction by the Belgian State, Prime Minister’s Office, Science Policy Programming (IUAP). The first author would like to acknowledge the partial support of the VW-foundation under Grant no. I/76938. W. Symens was a Research Assistant of the Fund for Scientific Research, Flanders (Belgium) (F.W.O.) during the period this research was conducted (1999–2003). The authors wish to thank Erik De Zeeuw and Diederik Douwen for their help in performing some of the experiments presented in this paper and Vincent Lampaert to put his set-up at their disposal. The support of guideway constructor Schneeberger is greatly acknowledged. The scientific responsibility is assumed by its authors.

References

- [1] J. Brändle, P. Eschmann, L. Hasbargen, K. Wiegand, Ball and Roller Bearings: Theory Design and Application, 3rd ed., Wiley, 1995.
- [2] R. Arnell, P. Davies, J. Halling, T. Whomes, Tribology, Macmillan Education Ltd., London, 1991.
- [3] B. Armstrong-Helouvry, Control of Machines with Friction, Kluwer Academic Publishers, 1991.
- [4] T. Prajogo, Experimental study of pre-rolling friction for motion-reversal error compensation on machine tool drive systems, Ph.D. Thesis, Division of Production Engineering, Machine Design and Automation, Katholieke Universiteit Leuven, Belgium, 1999.
- [5] F. Al-Bender, W. Symens, Dynamic characterisation of hysteresis elements in mechanical systems, Part I. Theoretical analysis, CHAOS Interdisc. J. Nonlinear Sci. (2005), in press.
- [6] J. Swevers, F. Al-Bender, C. Ganseman, T. Prajogo, An integrated friction model with improved presliding behaviour for accurate friction compensation, IEEE Trans. Autom. Contr. 45 (4) (2000) 675–686.
- [7] V. Lampaert, J. Swevers, F. Al-Bender, Modifications of the leuven integrated friction model structure, IEEE Trans. Autom. Contr. 47 (4) (2002) 683–687.
- [8] C. Hsieh, Y.-C. Pan, Dynamic behavior of modelling of the pre-sliding static friction, Wear 242 (2000) 1–17.
- [9] F. Al-Bender, V. Lampaert, J. Swevers, A novel generic model at asperity level for dry friction force dynamics, Tribol. Lett. 16 (1) (2004) 81–93.
- [10] V. Lampaert, F. Al-Bender, J. Swevers, Experimental characterisation of dry friction at low velocities on a developed tribometer setup for macroscopic measurements, Tribol. Lett. 16 (1) (2004) 95–105.
- [11] F. Al-Bender, V. Lampaert, J. Swevers, Modelling of dry sliding friction dynamics: from heuristic models to physical motivated models and back, CHAOS Interdisc. J. Nonlinear Sci. 14 (2) (2004) 446–460.
- [12] U. Parlitz, A. Hornstein, D. Engster, F. Al-Bender, V. Lampaert, T. Tjahjajawidodo, S. Fassois, D. Ri-zos, C. Wong, K. Worden, Identification of pre-sliding friction dynamics, CHAOS: Interdisc. J. Nonlinear Sci. 14 (2) (2004) 420–430.
- [13] J.J. Kalker, Three-dimensional elastic bodies in rolling contact, Kluwer, Dordrecht, 1990.
- [14] J.F. Cuttino, T.A. Dow, Contact between elastic bodies with an elliptic contact interface in torsion, Trans. ASME J. Appl. Mech. (64) (1997) 144–148.
- [15] J.F. Cuttino, T.A. Dow, B.F. Knight, Analytical and experimental identification of nonlinearities in a single-nut, preloaded ball screw, J. Mech. Des. (119) (1997) 15–19.

- [16] M.R. Lovell, M.M. Khonsari, R.D. Marangoni, The response of balls undergoing oscillatory motion: crossing from boundary to mixed lubrication regimes, *Trans. ASME J. Tribol.* 115 (1993) 261–266.
- [17] M.R. Lovell, M.M. Khonsari, R.D. Marangoni, Comparison of the low-speed frictional characteristics of silicon nitride and steel balls using conventional lubricants, *Trans. ASME J. Tribol.* 118 (1996) 43–51.
- [18] M.R. Lovell, M.M. Khonsari, R.D. Marangoni, Parameter identification of hysteresis friction for coated ball bearings based on three-dimensional FEM analysis, *Trans. ASME J. Tribol.* 119 (1997) 462–470.
- [19] M.R. Lovell, M.M. Khonsari, On the frictional characteristics of ball bearings coated with solid lubricants, *Trans. ASME J. Tribol.* 121 (1999) 761–767.
- [20] P.R. Dahl, A solid friction model, Technical report, The Aerospace Corporation, El Segundo, CA, 1968.
- [21] M.R. Hachkowski, L.D. Peterson, M.S. Lake, Load path management design theory for precision deployable joints, in: *Proceedings of the AIAA Structures, Structural Dynamics and Materials Conference*, 2000.
- [22] J.D. Hinkle, L.D. Peterson, P.A. Warren, Component level evaluation of a friction damper for submicron vibrations, in: *Proceedings of the 43rd Structures, Structural Dynamics and Materials Conference*, 2002.
- [23] L.D. Peterson, J.D. Hinkle, Microdynamic design requirements for large space structures, in: *Proceedings of the 44th Structures, Structural Dynamics and Materials Conference*, 2003.
- [24] L.D. Peterson, J.D. Hinkle, Microdynamic design and analysis of a structurally connected tpf interferometer, Final report for the period 1, Technical report, Structural Dynamics & Control Laboratory, Department of Aerospace Engineering Sciences, University of Colorado, May 2003–31 October 2003, CU-CAS-03-07, 2003.
- [25] J.A. Greenwood, K.L. Johnson, E. Matsubara, A surface roughness parameter in hertz contact, *Wear* 100 (1984) 47–57.
- [26] W. Symens, Motion and vibration control of mechatronic systems with variable configuration and local non-linearities, Ph.D. Thesis, Division of Production Engineering, Machine Design and Automation, Katholieke Universiteit Leuven, Belgium, 2004.
- [27] F. Al-Bender, Measurement of frictional hysteresis in ball-bearing guideways, preliminary report, Technical report, Division of Production Engineering, Machine Design and Automation, Katholieke Universiteit Leuven, Belgium, 2001RP39, 2001.
- [28] E. De Zeeuw, D. Douwen, Characterisation of friction hysteresis in linear guideways with application in position control, Master's Thesis, Mechanical Engineering Department, KUL, 2002 (in Dutch).
- [29] F. Al-Bender, Damping in re-circulating rolling-element guideways, Phase 1: overview and feasibility, Technical report, Division of Production Engineering, Machine design and Automation, Katholieke Universiteit Leuven, Belgium, 2000, 2000RP38.
- [30] I.D. Mayergoyz, *Mathematical Models of Hysteresis*, Springer-Verlag, New York, 1991.
- [31] W. Symens, F. Al-Bender, J. Swevers, H. Van Brussel, Dynamic characterisation of hysteresis elements in mechanical systems, in: *Proceedings of the American Control Conference*, May, 2002, pp. 4129–4134.
- [32] F. Al-Bender, W. Symens, J. Swevers, H. Van Brussel, Theoretical analysis of the dynamic behaviour of hysteresis elements in mechanical systems, *Int. J. Non-linear Mech.* 39 (2004) 1721–1735.
- [33] M. Goldfarb, N. Celanovic, A lumped parameter electromechanical model for describing the non-linear behaviour of piezoelectric actuators, *Trans. ASME J. Dyn. Syst. Measur. Contr.* 119 (1997) 478–485.
- [34] W. Symens, F. Al-Bender, J. Swevers, H. Van Brussel, Harmonic analysis of a mass subject to hysteretic friction: experimental validation, in: *Proceedings of the International Seminar on Modal Analysis*, September, 2002, pp. 1229–1240.
- [35] W. Symens, F. Al-Bender, Dynamic characterisation of hysteresis elements in mechanical systems, Part II: Experimental validation, *CHAOS: Interdisc. J. Nonlinear Sci.* (2005), in press.